

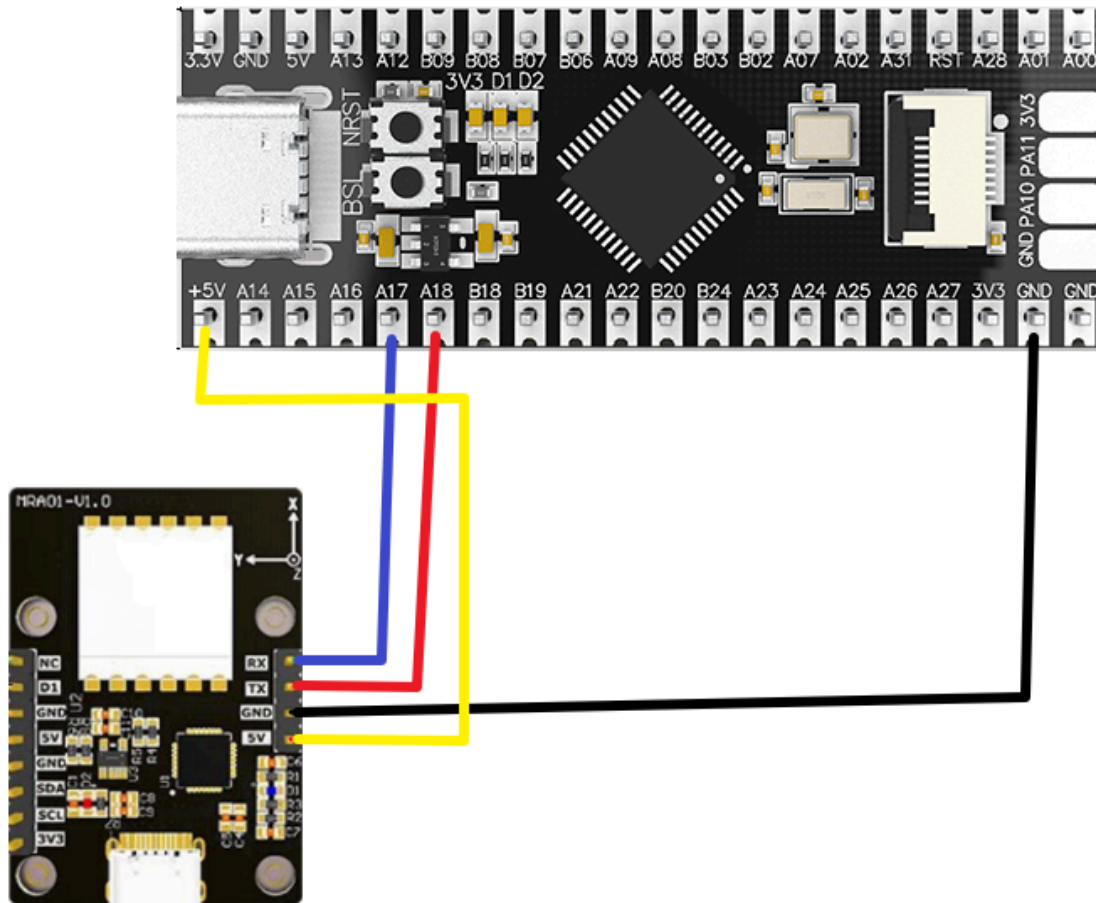
10-axis imu data acquisition

1. Learning objectives

Serial port prints the data of the imu module.

2. Hardware connection

MSPM0G3519 and ten-axis imu module pin connection



imu module	MSPM0G3519
RX	PA17
TX	PA18
5V	5V
GND	GND

3. Program description

This example sets the baud rate of the serial port printing to 9600 bps, and the baud rate of the IMU module connection to 9600 bps.

The screenshot displays the TI Studio configuration interface. On the left, a tree view shows the project configuration hierarchy: PROJECT CONFIGURATION (1/1), MSPM0 DRIVER LIBRARY, SYSTEM (9) including Board (1/1), DMA, GPIO, MATHACL, Configuration NVM, RTC, SYSCTL (1/1), SYSTICK (1/1), and WWDT; ANALOG (6) including ADC12, COMP, DAC12, GPAMP, OPA, and VREF; COMMUNICATIONS (6) including I2C, I2C - SMBUS, MCAN, SPI, and UART (2/4); and TIMERS (6) including TIMER - CAPTURE. The UART component is selected and highlighted. The main panel shows the configuration for 'MYUART' (selected from a list that also includes 'IMU_Uart'). The configuration is divided into sections: 'Quick Profiles' (UART Profiles set to 'Custom') and 'Basic Configuration'. The 'UART Initialization Configuration' section is expanded, showing settings for Clock Source (BUSCLK), Clock Divider (Divide by 1), Calculated Clock Source (32.00 MHz), Target Baud Rate (9600), Calculated Baud Rate (9600.24), Calculated Error (%) (0.0025), Word Length (8 bits), Parity (None), Stop Bits (One), and HW Flow Control (Disable HW flow control). The 'Target Baud Rate' field is highlighted with a red box.

PROJECT CONFIGURATION (1/1) ✓ +

MSPM0 DRIVER LIBRARY ...

SYSTEM (9)

- Board 1/1 ✓ +
- DMA +
- GPIO +
- MATHACL +
- Configuration NVM +
- RTC +
- SYSCTL 1/1 ✓ +
- SYSTICK 1/1 ✓ +
- WWDT +

ANALOG (6)

- ADC12 +
- COMP +
- DAC12 +
- GPAMP +
- OPA +
- VREF +

COMMUNICATIONS (6)

- I2C +
- I2C - SMBUS +
- MCAN +
- SPI +
- UART 2/4 ✓ +**
- UART - LIN +

TIMERS (6)

- TIMER - CAPTURE +

UART (2 of 4 Added) ⓘ

MYUART ✓

IMU_Uart ✓

Name MYUART

Selected Peripheral UART0

Quick Profiles ^

UART Profiles Custom v

Basic Configuration ^

UART Initialization Configuration ^

Clock Source BUSCLK v

Clock Divider Divide by 1 v

Calculated Clock Source 32.00 MHz

Target Baud Rate 9600

Calculated Baud Rate 9600.24 v

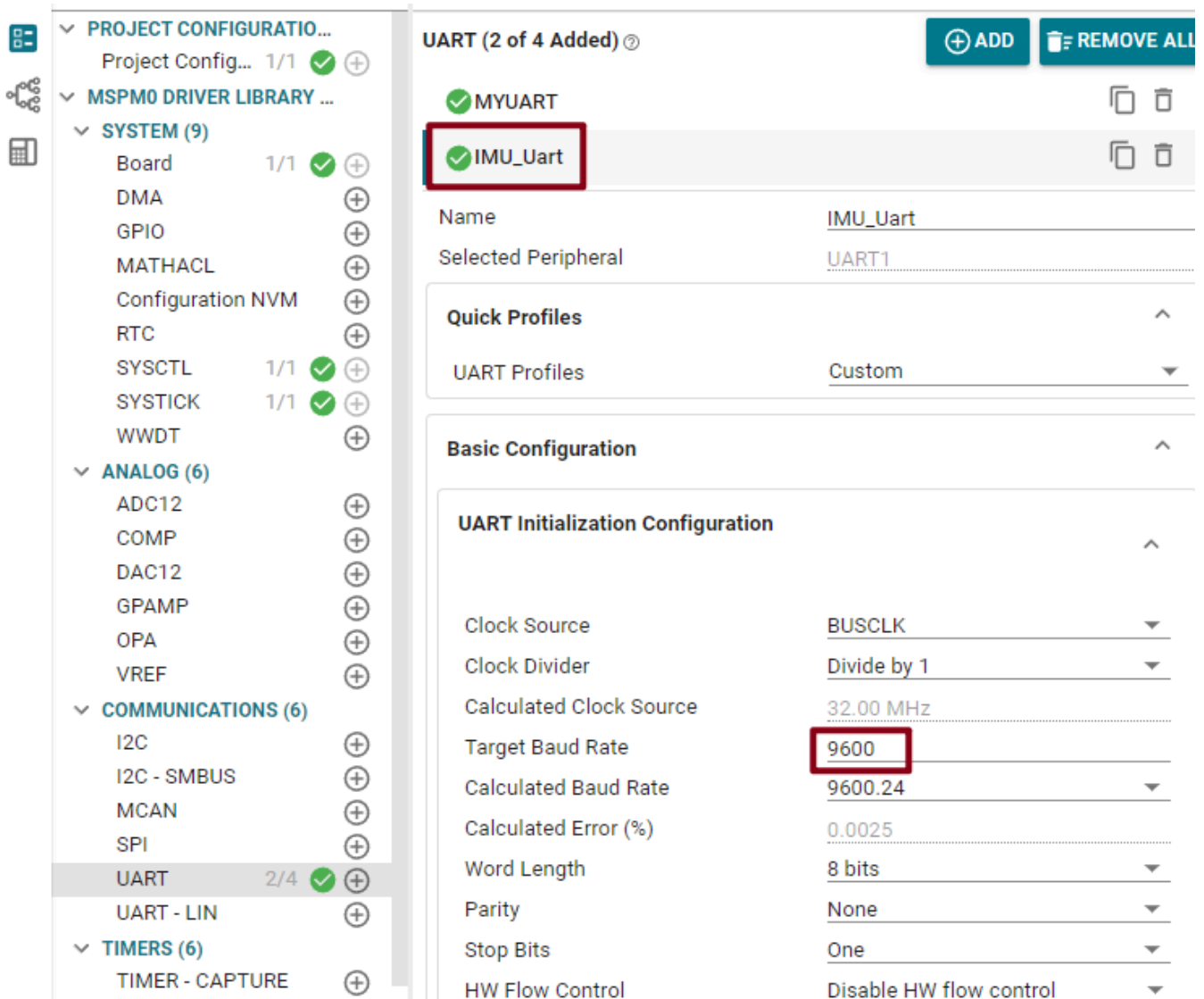
Calculated Error (%) 0.0025

Word Length 8 bits v

Parity None v

Stop Bits One v

HW Flow Control Disable HW flow control v



PROJECT CONFIGURATION...
Project Config... 1/1 ✓ (+)

MSPM0 DRIVER LIBRARY ...

SYSTEM (9)
Board 1/1 ✓ (+)
DMA (+)
GPIO (+)
MATHACL (+)
Configuration NVM (+)
RTC (+)
SYSCTL 1/1 ✓ (+)
SYSTICK 1/1 ✓ (+)
WWDT (+)

ANALOG (6)
ADC12 (+)
COMP (+)
DAC12 (+)
GPAMP (+)
OPA (+)
VREF (+)

COMMUNICATIONS (6)
I2C (+)
I2C - SMBUS (+)
MCAN (+)
SPI (+)
UART 2/4 ✓ (+)
UART - LIN (+)

TIMERS (6)
TIMER - CAPTURE (+)

UART (2 of 4 Added) ②

✓ MYUART (+) (-)
✓ IMU_Uart (+) (-)

Name IMU_Uart
Selected Peripheral UART1

Quick Profiles
UART Profiles Custom

Basic Configuration
UART Initialization Configuration

Clock Source BUSCLK
Clock Divider Divide by 1
Calculated Clock Source 32.00 MHz
Target Baud Rate **9600**
Calculated Baud Rate 9600.24
Calculated Error (%) 0.0025
Word Length 8 bits
Parity None
Stop Bits One
HW Flow Control Disable HW flow control

- bsp_imu.c

```
#include "bsp_imu.h"

volatile unsigned char imu_data = 0;

void Imu_init(void)
{
    NVIC_ClearPendingIRQ(IMU_Uart_INST_INT_IRQN); //清除串口中断标志 Clear the serial port interrupt flag
    NVIC_EnableIRQ(IMU_Uart_INST_INT_IRQN); //使能串口中断 Enable serial port interrupt
}

//串口发送单个字符
void uart1_send_char(char ch)
{
    //当串口1忙的时候等待，不忙的时候再发送传进来的字符
    //wait when serial port 1 is busy, and send the incoming characters when it is not busy
    while( DL_UART_isBusy(IMU_Uart_INST) == true );
}
```

```

//发送单个字符 Send a single character
DL_UART_Main_transmitData(IMU_Uart_INST, ch);
}
//串口发送字符 Send characters via serial port
void uart1_send_string(char* str)
{
    //当前字符串地址不在结尾 并且 字符串首地址不为空
    //The current string address is not at the end and the string's first address is not
empty
    while(*str!=0&&str!=0)
    {
        //发送字符串首地址中的字符，并且在发送完成之后首地址自增
        //Send the characters in the first address of the string, and the first address will
increment automatically after the sending is completed.
        uart1_send_char(*str++);
    }
}

//串口的中断服务函数 Serial port interrupt service function
void IMU_Uart_INST_IRQHandler(void)
{
    //如果产生了串口中断 If a serial port interrupt occurs
    switch( DL_UART_getPendingInterrupt(IMU_Uart_INST) )
    {
        case DL_UART_IIDX_RX://如果是接收中断 If it is a receive interrupt
            //接收发送过来的数据保存在变量中 The data received and sent is stored in the variable
            imu_data = DL_UART_Main_receiveData(IMU_Uart_INST);
            //将保存的数据再发送出去 Send the saved data again

//
            DueData(imu_data);//解析imu数据 Parsing imu data
//
            uart0_send_char(imu_data);
            break;

            default://其他的串口中断 Other serial port interrupts
                break;
    }
}
}

```

It mainly implements the function of receiving and initially processing IMU data based on UART, and uses functions such as DL_UART_isBusy and DL_UART_Main_transmitData to simplify the hardware operation of UART.

- imu_use.c

```

void DueData(uint8_t inputdata)
{
    if (inputdata == 0x55 && start == 0)
    {

```

```

        start = 1;
        data_length = 10;
        RxBuff[0] = 0x55;
        CheckSum = 0x55;
        return;
    }
    if(start == 1)
    {
        if(inputdata == 0x53) //因为速率太高，不能获取全部信息，该历程只获取姿态角度 Because the
speed is too high, it is not possible to obtain all the information. This process only
obtains the attitude angle
        {
            start = 2 ;
        }
        else
        {
            //防止其他数据出现0x53的有效数据 Prevent other data from showing valid data of 0x53
            memset(RxBuff,0,11);
            start = 0;
        }
    }

    if (start == 2)
    {
        CheckSum += inputdata; //校验码计算 会把校验位加上 The checksum calculation will add the
check digit
        RxBuff[11-data_length] = inputdata; //保存数据 Save data
        data_length = data_length - 1; //长度减一 Length minus one

        if (data_length == 0) //接收到完整的数据 Received complete data
        {
            CheckSum = (CheckSum-inputdata) & 0xff;
            start = 0; //清0 clear
            memcpy(RxBuffnew,RxBuff,11);

            CheckSumnew = CheckSum;
//            g_newflag = 1;
            memset(RxBuff,0,11);
            GetDataDeal();
        }
    }
}

```

The DueData function is used to process the data received from the IMU module through the serial port. It parses the data according to the module-specific protocol and prepares it for further processing.

```

void GetDataDeal(void)
{
    if(RxBuffnew[10] != CheckSumnew) //校验码不正确 The check code is incorrect
    {
        CheckSumnew = 0;
    }
}

```

```

        memset(RxBuffnew,0,11);
        return;
    }
    if(RxBuffnew[1] == 0x53) //姿态角度输出 Attitude angle output
    {
        angle[0] = (RxBuffnew[3] << 8 | RxBuffnew[2]) / 32768.0 * 180.0;
        angle[1] = (RxBuffnew[5] << 8 | RxBuffnew[4]) / 32768.0 * 180.0;
        angle[2] = (RxBuffnew[7] << 8 | RxBuffnew[6]) / 32768.0 * 180.0;
        printf("angle:x:%d y:%d z:%d \r\n" ,(int)angle[0],(int)angle[1],(int)angle[2]);

    }

    CheckSumnew = 0;
    memset(RxBuffnew,0,11);
}

```

The main purpose of the GetDataDeal function is to process the data received from the imu module. This data has been received and stored in the RxBuffnew array through the DueData function, and the checksum is stored in CheckSumnew. If the checksum is incorrect, the CheckSumnew and RxBuffnew arrays are cleared. If RxBuffnew[1] is 0x53, the attitude angle is output for a long time. The function will parse three angle values from the RxBuffnew array and print them out.

Note: The project source code must be placed in the SDK path for compilation.

For example, the path: D:\TI\M0_SDK\mspm0_sdk_1_30_00_03\TB6612

> 新加卷 (D:) > TI > M0_SDK > mspm0_sdk_1_30_00_03				
名称	修改日期	类型	大小	
1.TB6612	2024/7/22 18:59	文件夹		
2.AT8236	2024/7/22 19:47	文件夹		
3.Enconder	2024/7/23 10:36	文件夹		
4.Servo	2024/7/23 11:13	文件夹		
docs	2024/7/23 10:33	文件夹		
examples	2024/7/23 10:34	文件夹		
kernel	2024/7/23 10:37	文件夹		
source	2024/7/23 10:33	文件夹		
tools	2024/7/23 10:33	文件夹		
imports.mak	2024/1/25 11:45	MAK 文件	2 KB	
known_issues_FAQ.html	2024/1/25 11:42	Microsoft Edge ...	67 KB	
license_mspm0_sdk_1_30_00_03.txt	2024/1/25 11:42	文本文档	33 KB	
manifest_mspm0_sdk_1_30_00_03.html	2024/1/25 11:42	Microsoft Edge ...	113 KB	
mspm0sdk_1_30_00_03.log	2024/7/23 10:42	文本文档	5,237 KB	
release_notes_mspm0_sdk_1_30_00_0...	2024/1/25 11:42	Microsoft Edge ...	108 KB	
uninstall.dat	2024/7/23 10:39	DAT 文件	344 KB	
uninstall.exe	2024/7/23 10:39	应用程序	6,048 KB	